

GitHub Pages + Hugo (in English - external stuff)

How to create a website using Hugo and host it on GitHub

Introduction

In this post we will create a website using Hugo and host it as a GitHub Pages project and automate the whole process with Github Actions.

This is an update from an earlier post “Hugo in 10 Minutes”

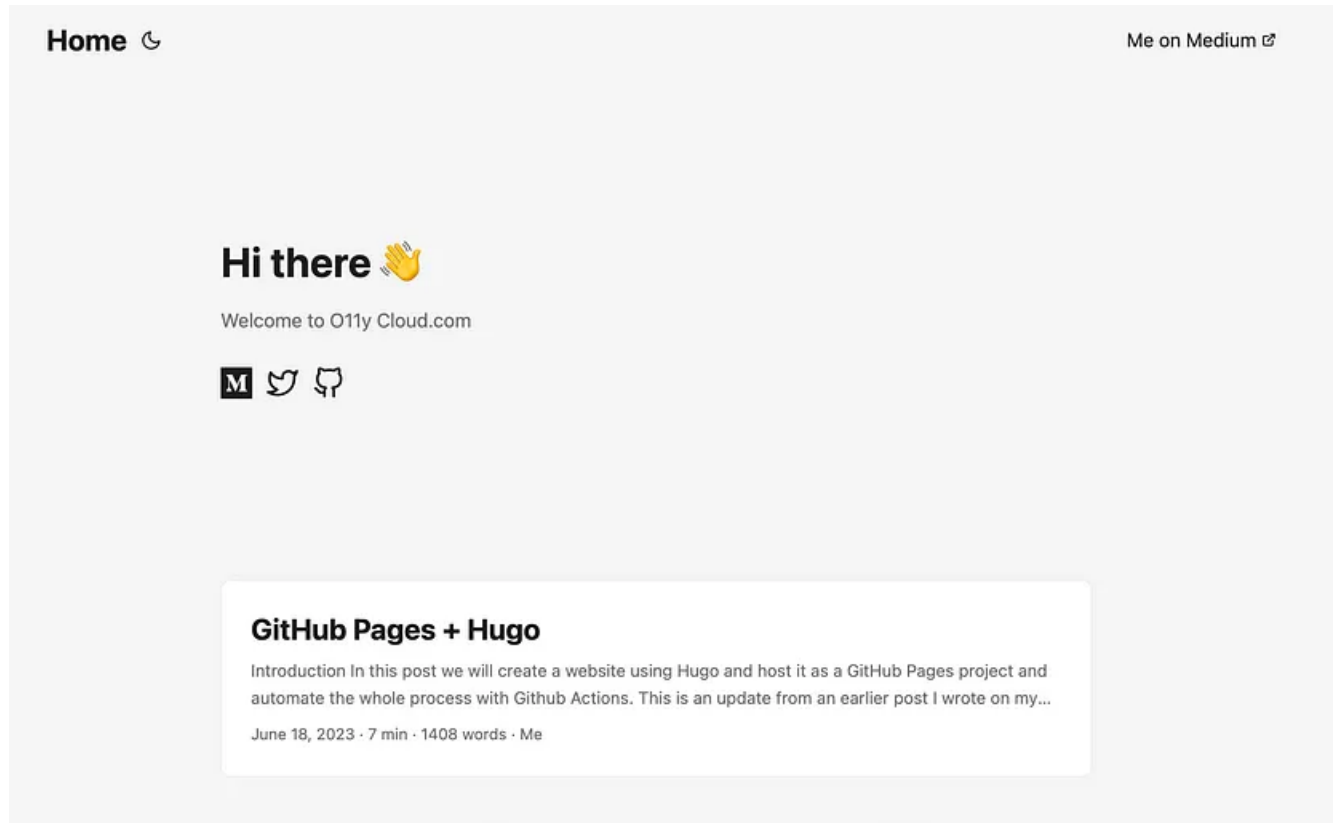


Рис. 1: 3f20db9e928521634b619292d27246bd.png

Prerequisites

- A GitHub account
- Hugo

GitHub Pages

GitHub Pages is a static site hosting service that takes HTML, CSS, and JavaScript files straight from a repository on GitHub, optionally runs the files through a build process, and publishes a website.

In addition to that, it's also free.

The requirements for using GitHub pages in our case are:

- To publish a user site, you must create a repository owned by your personal account that's named `<username>.github.io`.
- The source files for a project site are stored in the same repository as their project.
- You can only create one user or organization site for each account on GitHub.

There are also some limitations outlined here.

GitHub provides a good quick-start, so I won't go much in detail about that.

It usually comes down to these steps:

1. Create a repository and name it `username.github.io` as the repository name. (In my case, it will be **magsther.github.io**) The website will be published at `https://magsther.github.io/`
2. Clone the repository to your computer.

3. Cd into the folder
4. Add an index.html
5. Add, commit, and push your changes
6. Enable and configure Pages (in GitHub)
7. After a little while, you can visit `username.github.io` to view your new website.

Next step is to setup a GitHub Pages site with **Hugo**.

Hugo

*Hugo is a fast and modern static site generator written in Go, and designed to make website creation fun again. It is a popular **static site generator**, designed to provide an optimal viewing experience for your website's end users and an ideal writing experience for website authors.*

A static site generator is a tool that generates a full static HTML website based on raw data and a set of templates.

Hugo works on macOS, Linux, Windows and you can follow the installation instructions for your platform [here](#).

Once the installation is done, we can just go through the Quick-start guide, which more or less comes down to this:

These are the steps that we will do:

1. Create a Hugo project (site)
2. Choose a theme
3. Install and configure the theme
4. Customising the theme
5. Add content by creating a post
6. Run Hugo

Creating a Hugo Project (Site)

Start by cloning your repository to your computer.

```
git clone git@github.com:magsther/magsther.github.io.git
```

Cd into the folder and run the following command:

```
hugo new site ollycloud -f yml # creates the directory structure for your project
```

Just a few more steps and you're ready to go:

1. Download a theme into the same-named folder.
Choose a theme from <https://themes.gohugo.io/> or create your own with the "hugo new theme <THEMENAME>" command.
2. Perhaps you want to add some content. You can add single files with "hugo new <SECTIONNAME>/<FILENAME>.<FORMAT>".
3. Start the built-in live server via "hugo server". Visit <https://gohugo.io/> for quickstart.

Now we need to add a theme to your Hugo site, which is what we will do next.

Themes

Hugo provides a list of themes that you can use for your new website. If you are familiar with Wordpress then you will know how useful this is. Let's first to go the themes page and select a theme that you like.

I like the PaperMod theme, so I'll use that one.

Installing a Theme

Follow the instructions for the theme on how to **install** it. You can also watch a YouTube video where the author will guide you through the process.

When you select a theme it will give you instructions about the installation and configuration for that particular theme.

I cloned the repository and copied the PaperMod themes folder into the root themes folder

```
git clone https://github.com/adityatelange/hugo-PaperMod themes/PaperMod --depth=1
```

I also replaced the config.yml file (in the root module) and made some changes to it.

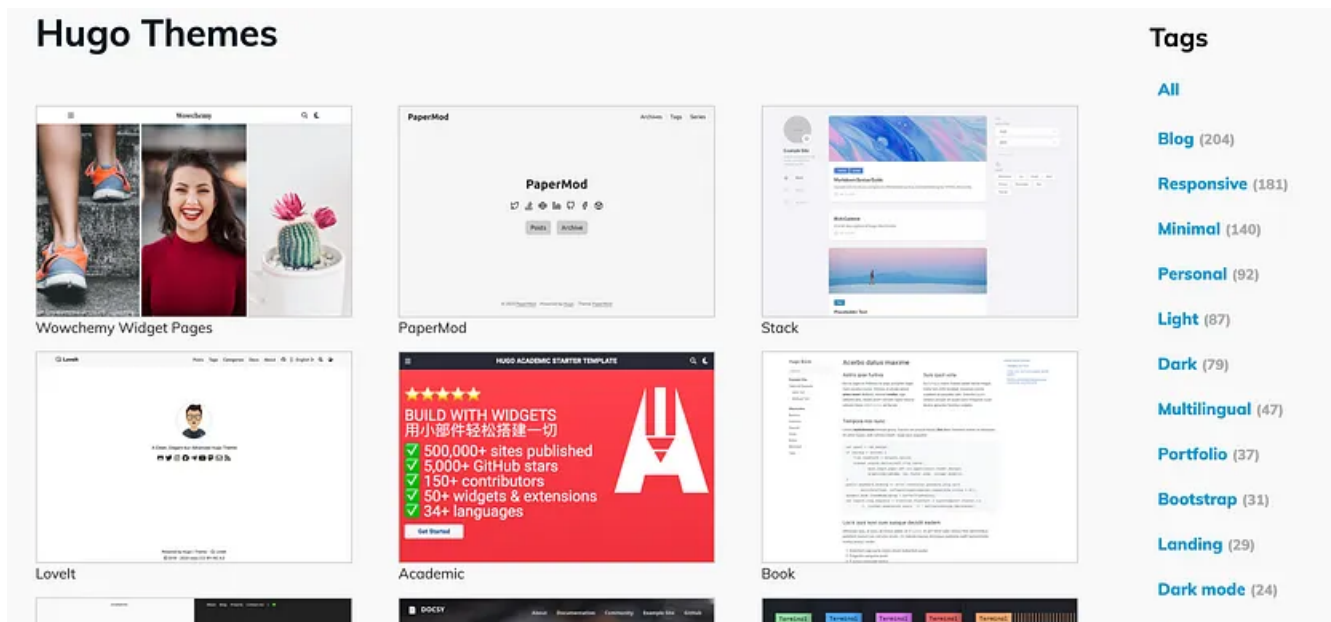


Рис. 2: 49dedc740f04e3bca1cbec3b9181cc67.png

```
baseURL: "http://magsther.github.io/"
title: OillyCloud.com
paginate: 5
theme: PaperMod
...
....
```

Run Hugo

To view the website locally, we can use Hugo's development server. Just type in `hugo server` at the prompt, and it will start building the website. At the bottom of the output we can find a link on how we can access the website.

Start building sites ...

	EN
Pages	7
Paginator pages	0
Non-page files	0
Static files	0
Processed images	0
Aliases	1
Sitemaps	1
Cleaned	0

Built in 32 ms

Environment: "development"

Serving pages from memory

Running in Fast Render Mode. For full rebuilds on change: `hugo server --disableFastRender`

Web Server is available at <http://localhost:1313/> (bind address 127.0.0.1)

Press Ctrl+C to stop

Open a browser and go to **<http://localhost:1313>**

Making changes on the “fly”

If you open a new terminal and modify some of the content in the `config.yml` file you will notice that website is updated automatically.

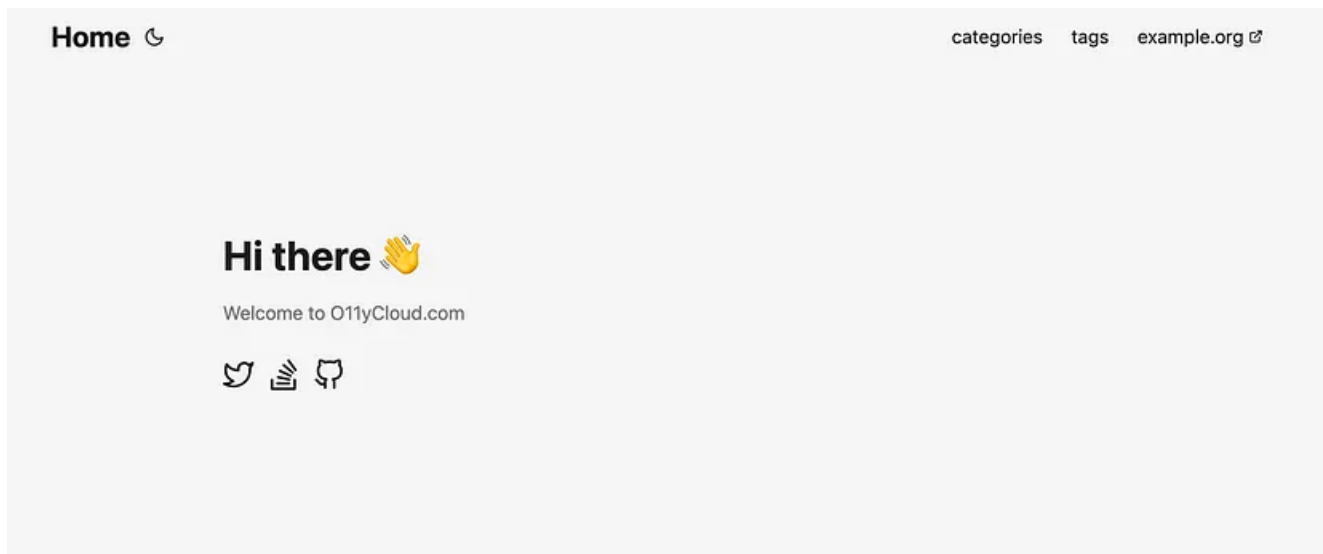


Рис. 3: 8b62c0ce1aefab2617a44dcec80f68b7.png

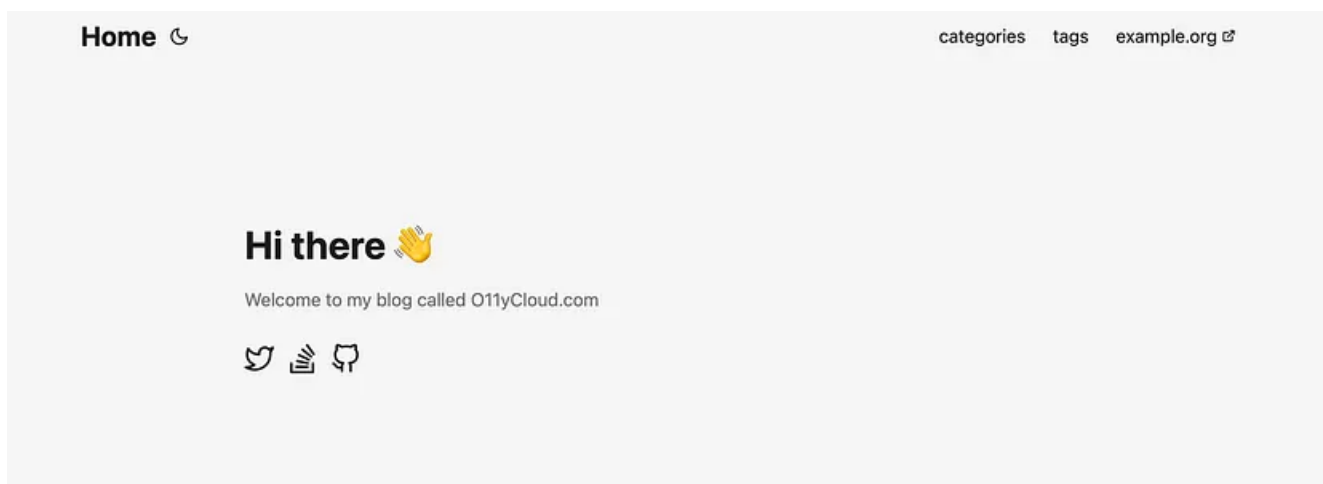


Рис. 4: ad8f6309b145ee9fb275a5c8411ee65d.png

This happened, because the Hugo process was still running in the other terminal and will build any changes that you make.

Change of config file detected, rebuilding site.

2023-06-11 23:15:55.936 +0200

Rebuilt in 63 ms

Tips, if you run `hugo server -D` then it will also included any posts that are in “draft” mode.

Adding content

To add content to your website, I first added a `post.md` file that was provided in the repository to the `archtypes` folder. You can think of that file as a template for all future posts. To create a new post, I typed in `hugo new posts/example.md`

This added an `example.md` file in the `content` folder.

You can now open this file with your editor to create your blog post.

If the Hugo process is still running and you are using the `-D` flag, then you will see that a new blog post is now added to your website.

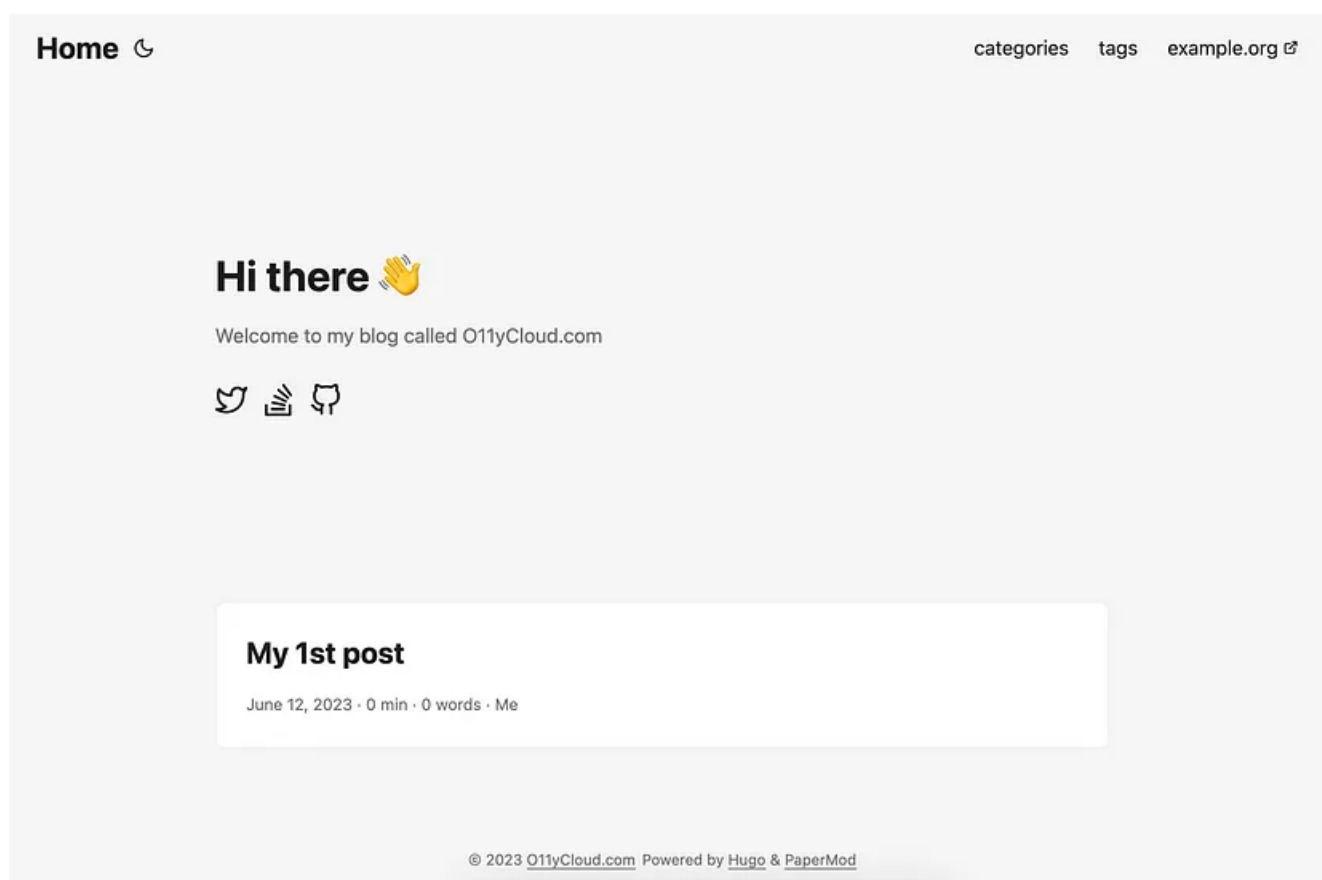


Рис. 5: 943eb2dee244c0d5f1620d99b4754769.png

Once you are done with the changes to your post, you can change `draft: true` to `draft: false` to have it deployed. This is depending on the theme you are using. If you are using the default theme, then Hugo does not publish draft content when you build the site

This should give you an understanding on how to create new blog posts.

Customizing Hugo

To further customize your new blog, just play around with the configuration file called `config.yml` located in the root folder.

The first things that you probably want to change are the `baseURL`, `title` and the `theme`

`baseURL: "https://magsther.github.io/"` #This value must begin with the protocol and end with
`title: O11y Cloud.com`

paginate: 5
theme: PaperMod

Publish your blog to GitHub Pages

When you are happy with how the website works on the local environment you can decide on how you want to host it. Here, we want to publish the blog to GitHub Pages.

To do that, we need to run `hugo` to generate the blog. That will create a `docs` folder that you will need to add an entry for in the `config.yml` file for.

After that, we add, commit and push them to the GitHub repository.

```
git add . && git commit -m 'new hugo blog' -a && git push
```

Next, we need to **enable** GitHub Pages, which we can do by going to our GitHub repository and from there click on `Pages`. Enable it by changing the Source from `None` to `Deploy from a branch`

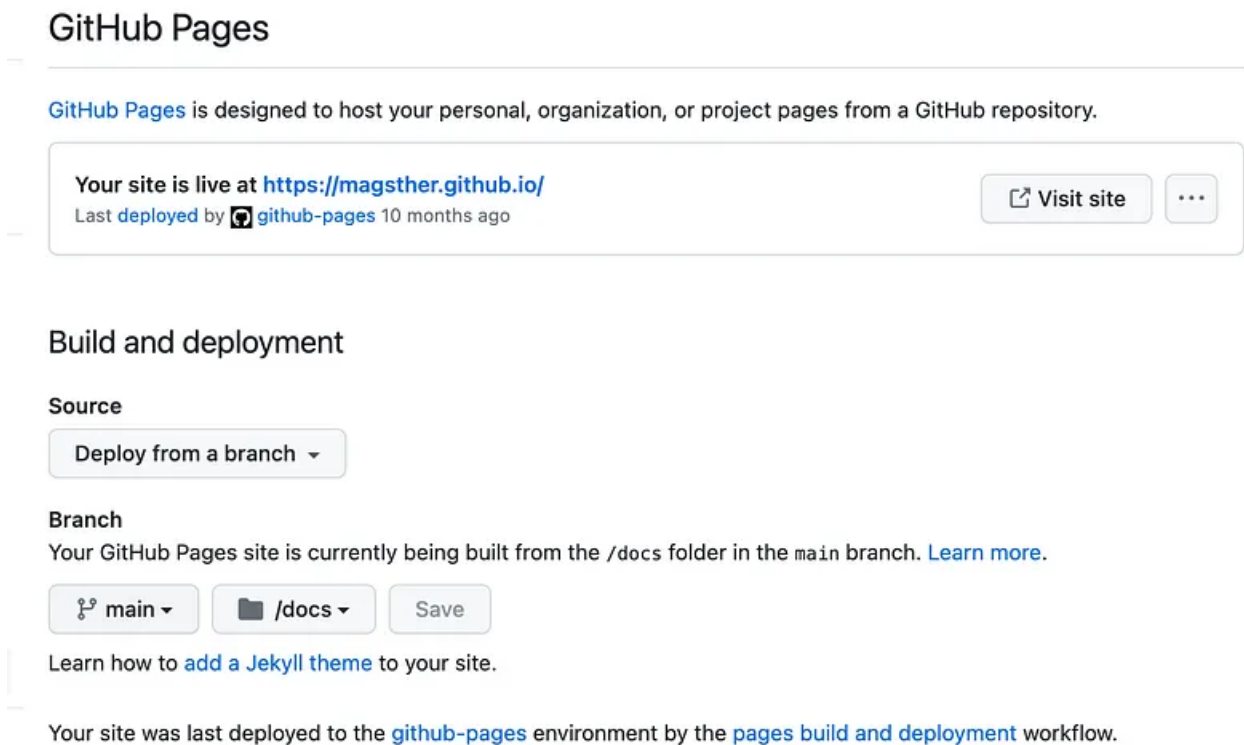


Рис. 6: 35ba15a47af1796ffa4eaf1c3e961727.png

Save your changes.

After a short while, you will see that your blog is now published at `https://username.github.io`

GitHub Actions

You may have noticed that a workflow was created for you in GitHub Actions.

If you click on the workflow, you can see a **summary** of the jobs that are part of this workflow.

In the future, whenever you push a change from your local repository, GitHub will rebuild your site and deploy the changes.

Updating your blog

To add a new post and publish it on your website, you can follow this workflow:

1. Create a new post with `hugo new posts/example2.md`
2. Build the site with `hugo`
3. Add, commit and push the changes to GitHub

```
git add . && git commit -m 'creating my second post' -a && git push
```

Actions

New workflow

All workflows

pages-build-deployment

Management

Caches

Deployments

All workflows

Showing runs from all workflows

Filter workflow runs

3 workflow runs

Event	Status	Branch	Actor
pages build and deployment	pages-build-deployment #3: by magsther	12 minutes ago	53s
pages build and deployment	pages-build-deployment #2: by magsther	13 minutes ago	53s
pages build and deployment	pages-build-deployment #1: by magsther	20 hours ago	54s

Рис. 7: ade90ff1b266ea55346a123eff4f6686.png

Summary

Jobs

build

report-build-status

deploy

Run details

Usage

Triggered via dynamic 15 minutes ago

Status

Total duration

Artifacts

magsther 758d7ef

Success

53s

1

pages-build-deployment

on: dynamic

build 19s

report-build-status 16s

deploy 9s

https://magsther.github.io/

Рис. 8: 8ba4b7bc5cd62844b1538fbf1ed1d7df.png

4. Watch the workflow in GitHub Actions
5. New posts are published at <https://username.github.io>